

A COUNTEREXAMPLE TO ERDŐS PROBLEM 108 VIA ARC GRAPHS

JENSEN KOHLMAYER AND LIAM KRUER

ABSTRACT. For every positive integer M , we construct a finite triangle-free graph G with $\chi(G) \geq M$ such that every ordinary subgraph of G containing no four-cycle is 6-colorable. Consequently, no chromatic threshold forces a subgraph of girth at least five and chromatic number at least seven. This gives a negative answer to Erdős problem 108.

The proof combines two ingredients. First, a four-cycle-free subgraph of an arc graph can be partitioned into a part controlled by a bounded-degree subgraph of the underlying graph and a forest. Second, a multipartite random construction gives graphs of large chromatic number for which every bounded-degree subgraph is 4-colorable, while retaining an acyclic orientation whose outdegree bound is independent of the prescribed degree cap. We give explicit integer part sizes and finite probability estimates. The main theorem and the negation of the pinned Formal Conjectures statement are also proved in Lean 4; the verification uses only the permitted standard axioms.

1. INTRODUCTION

The high-girth subgraph problem of Erdős and Hajnal asks whether, for every pair of integers $r \geq 4$ and $k \geq 2$, there is an integer $f(k, r)$ such that every graph of chromatic number at least $f(k, r)$ contains a subgraph of girth at least r and chromatic number at least k . The question appears as Problem 108 in Bloom’s catalogue [1] and is formalized in the Formal Conjectures library [2].

Throughout this paper, a subgraph may be obtained by deleting vertices and edges. In particular, “four-cycle-free” excludes a copy of C_4 as an ordinary subgraph; a four-cycle with chords is still forbidden.

Theorem 1.1. *For every integer $M \geq 1$, there is a finite nonempty triangle-free simple graph G such that $\chi(G) \geq M$ and every four-cycle-free subgraph H of G satisfies $\chi(H) \leq 6$.*

Corollary 1.2. *There is no integer f such that every graph of chromatic number at least f contains a subgraph of girth at least five and chromatic number at least seven. In particular, the assertion in Erdős problem 108 is false.*

Proof. Given a proposed threshold f , apply Theorem 1.1 with $M = \max\{1, f\}$. Every subgraph of girth at least five is four-cycle-free, so its chromatic number is at most six. $\square$

The construction starts with the multipartite random model used by Janzer, Steiner and Sudakov in their work on chromatic number and regular subgraphs [3]. We adapt its separated-part-size method to obtain a statement with two independent parameters: a chromatic parameter q and a degree cap K . The outdegree bound depends only on q . This permits us to choose K after fixing that bound and then apply an arc-graph argument. All estimates needed for this adaptation are proved here; the result is not used as a consequence of the regular-subgraph theorem.

Section 2 gives the arc-graph lemmas. Section 3 constructs the base graph, including the size estimates and both probability bounds. Section 4 proves Theorem 1.1. Section 5 describes the formal statements and the local verification record.

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Notation. All graphs used in the construction are finite and simple. We write $e_F(X)$ for the number of edges in $F[X]$, and $e_F(X, Y)$ for the number of edges between disjoint vertex sets X and Y . The maximum degree of a graph is denoted by Δ . A graph is 3-degenerate if every nonempty subgraph has a vertex of degree at most three; such a graph is 4-colorable. For an acyclic digraph D , let $U(D)$ be its underlying undirected graph. Forests are assigned infinite girth in the mathematical discussion.

2. ARC GRAPHS AND BOUNDED-DEGREE SELECTIONS

Definition 2.1. Let D be an acyclic orientation of a simple graph. Its *arc graph* $L(D)$ has the arcs of D as vertices, with two vertices adjacent precisely when the corresponding arcs concatenate. Thus (u, v) and (v, w) are adjacent. We orient this edge from (u, v) to (v, w) .

Fix a topological ordering of D . Along every oriented edge of $L(D)$, the tail of the corresponding arc strictly increases in this ordering. The displayed orientation of $L(D)$ is therefore acyclic.

Lemma 2.2. *The graph $L(D)$ is triangle-free.*

Proof. An acyclic orientation of a triangle has vertices a, b, c with edges $a \rightarrow b$, $b \rightarrow c$ and $a \rightarrow c$. If $a = (u, v)$ and $b = (v, w)$, the edge $b \rightarrow c$ makes the tail of c equal to w , whereas $a \rightarrow c$ makes it equal to v . Hence $v = w$, contradicting the fact that b is an arc of a simple graph. $\square$

Lemma 2.3. *If $L(D)$ is t -colorable, then $U(D)$ is 2^t -colorable. In particular,*

$$\chi(U(D)) \leq 2^{\chi(L(D))}. \quad (1)$$

Proof. Fix a proper coloring of $L(D)$ with colors in $\{1, \dots, t\}$. Assign to each vertex v of D the set of colors occurring on arcs directed out of v . If (u, v) has color c , the set at u contains c , but the set at v does not: every arc directed out of v is adjacent to (u, v) . Adjacent vertices of $U(D)$ therefore receive distinct subsets of the t -element color set. $\square$

Lemma 2.4 (Arc selection). *Suppose that D is acyclic, has maximum outdegree at most d , and every subgraph of $U(D)$ with maximum degree at most*

$$K_d = d + \binom{d}{2} \quad (2)$$

is a -colorable. Then every four-cycle-free subgraph of $L(D)$ is $(a + 2)$ -colorable.

Proof. Let $H \subseteq L(D)$ be four-cycle-free, and restrict the natural orientation to H . Partition its vertex set into S and T , where S consists of the arc vertices having at least two outgoing neighbors in H , and T consists of the remaining arc vertices.

Fix a vertex v of D . For every incoming arc (u, v) belonging to S , choose an unordered pair of its outgoing neighbors in H . These neighbors are arcs of the form (v, w) . Two distinct incoming arcs cannot receive the same pair: the two incoming arcs and the two chosen outgoing arcs, together with the four relevant edges of H , would form a four-cycle. Consequently, at most

$$\binom{d_D^+(v)}{2} \leq \binom{d}{2}$$

incoming arcs at v belong to S .

Let D_S retain precisely the arcs of D corresponding to vertices in S . Its outdegrees are at most d and its indegrees are at most $\binom{d}{2}$. Hence $\Delta(U(D_S)) \leq K_d$. By hypothesis there is a proper a -coloring c of $U(D_S)$. Color an arc vertex $(u, v) \in S$ by $c(u)$. If (u, v) and (v, w) are adjacent in $H[S]$, the edge uv belongs to $U(D_S)$, so these two arc vertices receive different colors.

The inherited orientation of $H[T]$ is acyclic and every vertex has outdegree at most one. Such a graph is a forest: any undirected cycle, with an acyclic orientation, has a source on that cycle and therefore a vertex with two outgoing edges. Color $H[T]$ with two colors disjoint from the first palette. This gives an $(a + 2)$ -coloring of H . $\square$

Remark 2.5. The partition and the selected graph D_S depend on the edge-deleted subgraph H . The argument does not require H to be induced. Also, the indegree bound is asserted only for D_S , not for the original digraph D .

3. THE MULTIPARTITE BASE GRAPH

Theorem 3.1. *For every pair of integers $q \geq 2$ and $K \geq 4$, there is a finite simple graph F with an acyclic orientation D such that:*

- (1) $\chi(F) > 2q$;
- (2) *the maximum outdegree of D is at most $64q^2 - 1$;*
- (3) *every subgraph $J \subseteq F$ with $\Delta(J) \leq K$ is 3-degenerate, and hence 4-colorable.*

In particular, the outdegree bound in (2) is independent of K .

3.1. The random model and explicit sizes. Put

$$C = 64q^2, \quad h = 2(K + 1), \quad A = 16CK(K + 1), \quad b = h + 2. \quad (3)$$

Choose the following positive integer R , and then define B and the part sizes:

$$R = (16C)^h 3^{2h+1} A + 3A + 8C + 2, \quad (4)$$

$$B = b^{C+1}, \quad n_i = R^{B-b^{i+1}} \quad (0 \leq i < C). \quad (5)$$

Take disjoint sets $V_0, \dots, V_{C-1}$ with $|V_i| = n_i$ and no edges inside a part. For every $v \in V_i$ and every $j > i$, independently choose a uniformly distributed vertex $w(v, j) \in V_j$ and insert the edge $vw(v, j)$. Orient each edge from its earlier part to its later part. This orientation is acyclic and has maximum outdegree at most $C - 1$.

Write $N = \sum_i n_i$ and $N_0 = R^B$. The following elementary estimates are the reason for the choice of exponents.

Lemma 3.2. *The sizes in (5) satisfy*

$$n_i \geq R \geq 8C, \quad Rn_j \leq n_i \quad (i < j), \quad (6)$$

$$N \leq Cn_0 \leq An_0, \quad N \leq N_0, \quad (7)$$

$$\sum_{j>i} n_j \leq n_i, \quad \sum_{j \geq i} n_j \leq 2n_i. \quad (8)$$

For $1 \leq i < C$, put $L = n_{i-1}$. For every integer x with $1 \leq x \leq An_i$, one also has

$$3x \leq L, \quad \frac{3^{2h+1} N_0^h x}{L^{h+1}} \leq (16C)^{-h}. \quad (9)$$

Proof. Since $b > 1$ and $i < C$, the exponent $B - b^{i+1}$ is a positive integer. For $i < j$ the two exponents differ by at least one, which proves (6). The sizes are decreasing. Since $R \geq C$ and each $n_i \leq R^{B-1}$, we obtain

$$N \leq CR^{B-1} \leq R^B = N_0.$$

The other inequalities in (7) use $A \geq C$. Moreover, $Rn_j \leq n_i$ for each $j > i$, so $R \sum_{j>i} n_j \leq Cn_i \leq Rn_i$, proving (8).

Fix $i \geq 1$. Since $R \geq 3A$, the successive-size inequality gives $3x \leq 3An_i \leq Rn_i \leq n_{i-1} = L$. Next, the identity $b = h + 2$ gives

$$(h + 1)(B - b^i) - hB - (B - b^{i+1}) = b^i.$$

Thus $L^{h+1} = R^{b^i} N_0^h n_i \geq R N_0^h n_i$. It follows that

$$\frac{3^{2h+1} N_0^h x}{L^{h+1}} \leq \frac{3^{2h+1} A}{R} \leq (16C)^{-h},$$

where the last inequality is immediate from (4). $\square$

3.2. The deterministic cut. Consider the following local sparsity condition on a realization of F :

$$\begin{aligned} X \subseteq V_0 \cup \dots \cup V_{i-1}, \quad 0 < |X| \leq An_i \\ \implies e_F(X) < (1 + 1/h)|X|. \end{aligned} \tag{10}$$

For $i = 0$ the condition is vacuous.

Lemma 3.3. *If (10) holds for every i , then every subgraph of F with maximum degree at most K is 3-degenerate.*

Proof. Suppose that $J \subseteq F$ has minimum degree at least four and maximum degree at most K , and let $s = |V(J)| > 0$. By (7), there is an index i with $s \leq An_i$. Choose the largest such index. Partition $V(J)$ into X, Y, Z , according as the part index is less than, equal to, or greater than i . If $i = C - 1$, then $Z = \emptyset$. Otherwise, maximality gives $An_{i+1} < s$, and the decreasing sizes give

$$|Z| \leq Cn_{i+1} < \frac{Cs}{A} = \frac{s}{16K(K+1)}.$$

In either case we may use the weaker bound

$$|Z| < \frac{2s}{16K(K+1)}. \tag{11}$$

Each vertex of X has at most one neighbor in V_i , while the degree cap gives at most $K|Y|$ edges incident with Y . Hence

$$e_J(X, Y) \leq |X|, \quad e_J(X, Y) \leq K|Y|.$$

Taking the weighted average with weights $K/(K+1)$ and $1/(K+1)$ yields

$$e_J(X, Y) \leq \frac{K}{K+1}(|X| + |Y|) \leq \frac{K}{K+1}s. \tag{12}$$

The set Y is independent. At most $K|Z|$ edges have an endpoint in Z , and the minimum-degree assumption gives $e(J) \geq 2s$. Therefore

$$\begin{aligned} e_J(X) &\geq 2s - e_J(X, Y) - K|Z| \\ &> \left(2 - \frac{K}{K+1} - \frac{1}{8(K+1)}\right)s \\ &= \left(1 + \frac{7}{8(K+1)}\right)s > (1 + 1/h)|X|. \end{aligned} \tag{13}$$

In particular X is nonempty, and $|X| \leq s \leq An_i$. Since $e_J(X) \leq e_F(X)$, (13) contradicts (10).

Thus there is no subgraph with all degrees between four and K . Every subgraph of a graph of maximum degree at most K still has maximum degree at most K , so every such graph is 3-degenerate. $\square$

3.3. The probability of a dense earlier set.

Lemma 3.4. *The probability that (10) fails for some i is at most $1/8$.*

Proof. Fix $i \geq 1$ and $1 \leq x \leq An_i$, and put

$$L = n_{i-1}, \quad m = \lceil (1 + 1/h)x \rceil, \quad \delta = \frac{1}{16C}.$$

For a specified collection of m distinct possible edges among the earlier parts, either two edges demand different choices from the same pair (v, j) , in which case the event is impossible, or they impose m independent choices. Every target part has size at least L , so the probability that all the specified edges occur is at most L^{-m} .

There are at most $\binom{N_0}{x}$ possible sets X of size x , and at most $\binom{x^2}{m}$ choices of the required edges. Thus the probability that an earlier set of size x violates (10) is at most

$$\binom{N_0}{x} \binom{x^2}{m} L^{-m} \leq \left(\frac{3N_0}{x} \right)^x \left(\frac{3x}{L} \right)^m. \quad (14)$$

Here we used $\binom{n}{k} \leq (en/k)^k \leq (3n/k)^k$ for $k \geq 1$, together with $m \geq x$. If the prescribed number of edges is impossible, the same estimate holds with probability zero.

Set $a = 3N_0/x$ and $z = 3x/L$. By Lemma 3.2, $0 < z \leq 1$. Since $hm \geq (h+1)x$, raising the right side of (14) to the integer power h gives

$$\begin{aligned} (a^x z^m)^h &\leq (a^h z^{h+1})^x \\ &= \left(\frac{3^{2h+1} N_0^h x}{L^{h+1}} \right)^x \leq \delta^{hx}. \end{aligned}$$

It follows that (14) is at most δ^x . Since $\delta \leq 1/2$, summing over the possible positive sizes gives a failure probability at most

$$\sum_{x \geq 1} \delta^x = \frac{\delta}{1-\delta} \leq 2\delta = \frac{1}{8C}$$

for a fixed i . Summing over at most C indices proves the result. $\square$

3.4. The probability of a heavy independent set. Assign weight $1/n_i$ to each vertex of V_i . For a vertex set I , write

$$p_i(I) = \frac{|I \cap V_i|}{n_i}, \quad w(I) = \sum_{i=0}^{C-1} p_i(I). \quad (15)$$

The total vertex weight is C .

Lemma 3.5. *The probability that F has an independent set I with $w(I) \geq C/(2q)$ is at most $1/8$.*

Proof. Suppose that $w(I) \geq C/(2q)$. Choose the first index i with $p_i(I) \geq 1/(4q)$. Such an index exists. Earlier parts contribute less than $C/(4q)$ and $p_i(I) \leq 1$, so

$$\sum_{j>i} p_j(I) > \frac{C}{4q} - 1 = 16q - 1 \geq 15q. \quad (16)$$

The part of I in indices at least i is still independent and preserves both inequalities. We may therefore count only sets in this suffix.

Fix a suffix set I satisfying these inequalities, and put $t = |I \cap V_i| \geq n_i/(4q)$. For I to be independent, all choices from its t vertices in V_i to each later part V_j must avoid $I \cap V_j$. These choices are independent, so

$$\mathbb{P}(I \text{ is independent}) \leq \prod_{j>i} (1 - p_j(I))^t \leq \exp \left(-t \sum_{j>i} p_j(I) \right) \leq e^{-3n_i}. \quad (17)$$

Indeed, (16) and $t \geq n_i/(4q)$ give an exponent of absolute value at least $15n_i/4$, which is more than sufficient.

By (8), the entire suffix has at most $2n_i$ vertices. There are therefore at most $2^{2n_i} \leq e^{2n_i}$ possible suffix sets. The union bound and (17) give probability at most e^{-n_i} for a witness with a fixed first heavy index i . Finally, $n_i \geq 8C$ and $e^{n_i} \geq 1 + n_i$ imply

$$e^{-n_i} \leq \frac{1}{8C}.$$

Summing over all i proves the lemma. $\square$

Proof of Theorem 3.1. Lemmas 3.4 and 3.5 bound the two failure probabilities by $1/8$ each. Their sum is less than one, so there is a realization satisfying (10) in which every independent set has weight less than $C/(2q)$. Lemma 3.3 proves (3), and the orientation of the construction proves (2).

If F were $2q$ -colorable, its color classes would be independent sets of total weight strictly less than $2q \cdot C/(2q) = C$. But the color classes partition a vertex set of weight C . This contradiction proves $\chi(F) > 2q$, completing (1). $\square$

4. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.1. Fix $M \geq 1$ and put

$$q = 2^M + 2, \quad C = 64q^2, \quad d = C - 1, \quad K = d + \binom{d}{2}. \quad (18)$$

These parameters satisfy $q \geq 2$ and $K \geq 4$. Apply Theorem 3.1 with this pair (q, K) to obtain $F = U(D)$. Every subgraph of F with maximum degree at most K is 4-colorable, and the maximum outdegree of D is at most d . The choice of K in (18) is legitimate because the outdegree bound in Theorem 3.1 is independent of K .

Set $G = L(D)$. Lemma 2.4, with $a = 4$, shows that every four-cycle-free subgraph of G is 6-colorable. Lemma 2.2 shows that G is triangle-free. If $\chi(G) \leq M$, then Lemma 2.3 gives

$$\chi(F) \leq 2^{\chi(G)} \leq 2^M < 2q,$$

contrary to Theorem 3.1. Thus in fact $\chi(G) > M$. The graph G is finite, and this inequality also ensures that it is nonempty. $\square$

Remark 4.1. List the vertices of D in a topological order. The map sending an arc to the ordered pair of its endpoint indices identifies $L(D)$ with an induced subgraph of the canonical shift graph on ordered pairs (i, j) with $i < j$, where (i, j) is adjacent to (j, k) . This description is not needed for the proof of Corollary 1.2.

5. LEAN FORMALIZATION AND VERIFICATION

The accompanying file `Erdos108.lean` formalizes the construction and the two conclusions used here. The theorem `Bounty.paper_main` asserts, for each $M \in \mathbb{N}$, the existence of a finite nonempty simple graph that is triangle-free, has chromatic number at least M , and has every four-cycle-free ordinary subgraph 6-colorable. The theorem `Bounty.target` proves the negation of the specified `Erdos108.erdos_108` type.

The formalization represents vertices of the multipartite model as pairs (i, x) with $i < C$ and $x < n_i$, in lexicographic order. A sample is a finite dependent product of choices; unused choices into earlier parts are retained so that the sample space is rectangular. Uniform probabilities are finite-set cardinalities divided by the cardinality of the sample space. Thus the existence argument uses exact finite counting, with real-valued estimates proved in Lean, rather than simulation or unverified numerical evaluation.

The graph-theoretic statements are formulated in terms of colorability. The predicate `FourCycleFree` excludes two distinct vertices having two distinct common neighbors; in a simple graph this is exactly the absence of an ordinary four-cycle. Subgraphs are handled through `SimpleGraph.Subgraph`, including both vertex and edge deletions. The main correspondences are listed in Table 1.

The Lean proof establishes the 4-colorability conclusion of Theorem 3.1 directly by finite-set induction using the low-degree vertex supplied by the cut argument. No separate library definition of degeneracy is required. Remark 4.1 is a mathematical observation and is not a separate formalized assertion.

The library chromatic number takes values in the extended natural numbers. Its natural-valued girth is zero for an acyclic graph, whereas we use infinite girth for forests in this paper. The proof only uses the implication from library girth at least five to four-cycle-freeness. The stronger four-cycle-free conclusion of Theorem 1.1 also covers forests, so this convention does not affect the counterexample. The final family is transported by `ULift` to the arbitrary universe quantified over by the target statement.

TABLE 1. Selected correspondences with the accompanying Lean file. All declaration names lie in namespace Bounty.

Mathematical statement	Lean declaration
Lemma 2.2	<code>arcGraph_triangle_free</code>
Lemma 2.3	<code>base_colorable_of_arc_colorable</code>
Lemma 2.4	<code>arc_subgraph_colorable_bound</code>
Lemma 3.2	<code>model_sizes_exist</code>
Lemma 3.3, coloring conclusion	<code>colorable_four_of_sparse_cuts</code>
Lemma 3.4	<code>model_badSparse_probability_le</code>
Lemma 3.5	<code>model_badColor_probability_le</code>
Theorem 3.1, needed conclusions	<code>model_base_exists</code>
Theorem 1.1	<code>paper_main</code>
Corollary 1.2, pinned target	<code>target</code>

Pinned environment and scope of verification. The local checks use Lean 4.33.1 [4], Mathlib revision 0df444a360ea, and the Formal Conjectures task pinned to revision 8432eac99811. The full revisions and artifact digest appear in the accompanying verification record. The checked target has the form

```
theorem target :
  ¬ (fcTypeOfName% "Erdos108.erdos_108") := by
  exact counterexample_of_family counterexample_family
```

The task’s type extractor is used only to obtain the statement. The imported conjecture is not used as a proof. The local axiom audit for both `Bounty.target` and `Bounty.paper_main` reports exactly

`propext,    Classical.choice,    Quot.sound.`

The delivered submission file contains no imports or outer namespace: the task supplies those in its trusted wrapper. The entire delivered file passes the task’s static submission policy; after adding that wrapper, it compiles and its target type matches the pinned manifest. The file contains no admitted proof or added axiom. These statements report local formal verification. They do not report an online submission decision or independent peer review.

6. CONCLUDING REMARKS

The obstruction is uniform: the bound six does not depend on the chromatic number of the host graph. Hence the same family excludes a threshold for every girth parameter $r \geq 5$ at chromatic parameter $k = 7$. The proof gives no claim that six is the smallest possible uniform bound.

The sizes in (4)–(5) were chosen to simplify the proof and its formalization. They specify a finite construction, but are far too large to make enumeration of the required realizations practical. The conclusion is an existence theorem; small computational examples are not used to certify the probability estimates.

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PURDUE UNIVERSITY, WEST LAFAYETTE, INDIANA, USA
Email address: jenwin@purdue.edu

PURDUE UNIVERSITY, WEST LAFAYETTE, INDIANA, USA
Email address: kruerl@purdue.edu